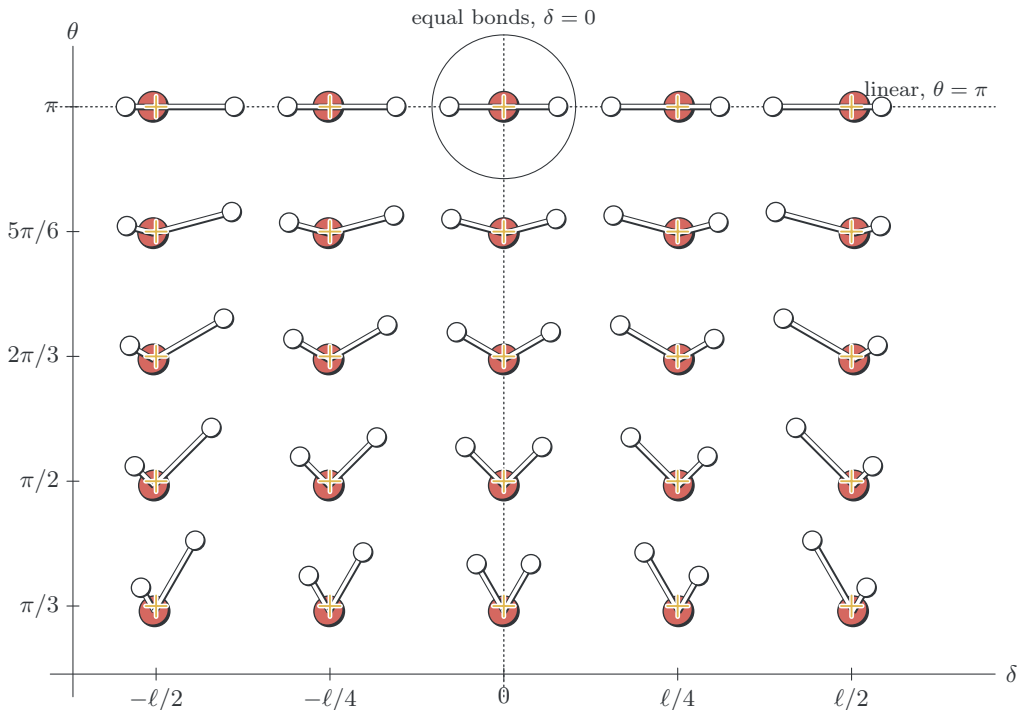
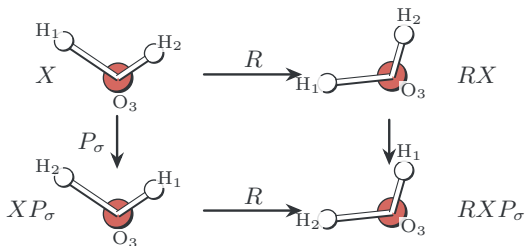




page position encodes  $(\delta, \theta)$  with  $r_1 = \ell + \delta$ ,  $r_2 = \ell - \delta$ ; the gold cross is each configuration's own origin, its mass centre,  $0.06 \text{ \AA}$  from the oxygen in the shape at upper left; H<sub>1</sub> left, H<sub>2</sub> right; orientation suppressed



$\sigma = (12)$  swaps the columns;  $R = R_z(40^\circ)$  multiplies every column;  
 $(RX)P_\sigma = R(XP_\sigma)$

$$X = \begin{pmatrix} -0.93 & 0.65 & 0.02 \\ 0.58 & 0.37 & -0.06 \\ 0 & 0 & 0 \end{pmatrix} \text{ \AA}$$

$$X\mathbf{m} = 0, \mathbf{m} = (1.008, 1.008, 15.995)^\top \text{ u}$$